

List of puzzles

1. Crossing the bridge in 17 seconds.

Source: <http://puzzles.nigelcoldwell.co.uk/twentyfive.htm>

Adam, Bob, Clair and Dave are out walking: They come to rickety old wooden bridge. The bridge is weak and only able to carry the weight of two of them at a time. Because they are in a rush and the light is fading, they must cross in the minimum time possible and must carry a torch (flashlight,) on each crossing.

They only have one torch and it can't be thrown. Because of their different fitness levels and some minor injuries, they can all cross at different speeds. Adam can cross in 1 minute, Bob in 2 minutes, Clair in 5 minutes and Dave in 10 minutes.

Adam, the brains of the group thinks for a moment and declares that the crossing can be completed in 17 minutes. There is no trick. How is this done?

2. measuring exactly four litres of water.

Source: <https://javarevisited.blogspot.com/2013/04/top-10-puzzles-riddles-logical-programming-question-lateral-thinking-interview.html#ixzz4cSiD4QMG>

You have a 5-liter Jug and a 3-liter Jug and an unlimited supply of water. You need to measure exactly 4-litre of water but there is no measuring instrument or cup. Also, Jugs are oddly shaped and don't contain any marks.

3. Number of squares in a chess board.

Source: <https://www.teachingideas.co.uk/problem-solving/how-many-squares-on-a-chessboard>

How many squares in a chess board?

4. Write code for the FIZZBUZZ problem.

Source: <http://wiki.c2.com/?FizzBuzzTest>

Write a program that prints the numbers from 1 to 100. But for multiples of three print "Fizz" instead of the number and for the multiples of five print "Buzz". For numbers which are multiples of both three and five print "FizzBuzz".

5. Suicide circle.

Source: <https://www.thesun.co.uk/news/2196722/could-you-survive-the-gruesome-suicide-circle-brain-teaser-made-infamous-by-jewish-soldiers-nearly-2000-years-ago/>

The 41 rebel soldiers made a suicide pact, deciding they would quickly form a circle. The first man chosen would then kill the man to his left, and this process would be repeated until there was only one left standing. That man would then kill himself, meaning none could be captured. The troops were later forced to put their plan into action as Romans closed in on them during the Siege of Yodfat. Among them was historian Flavius Josephus, who wanted to survive the ordeal but did not want to lose the respect of his comrades by turning himself over to the Romans.

6. 25 horses puzzle.

Source: <https://mindyourdecisions.com/blog/2017/05/11/can-you-solve-the-25-horses-puzzle-google-interview-question/>

There are 25 horses among which you need to find out the fastest 3 horses. You can conduct race among at most 5 to find out their relative speed. At no point you can find out the actual speed of the horse in a race. Find out the minimum number of races which are required to get the top 3 horses.

7. 1000 coins and 10 bags.

Source: <https://www.geeksforgeeks.org/puzzle-11-1000-coins-and-10-bags/?ref=lbp>

A dealer has 1000 coins and 10 bags. He must divide the coins over the ten bags so that he can make any number of coins simply by handing over a few bags. How must divide his money into the ten bags?

8. Two marbles and 100 floors.

Source: [algorithm - Two marbles and a 100 story building - Stack Overflow](https://stackoverflow.com/questions/1176052/two-marbles-and-a-100-story-building)

You are given two marbles and told that they will break when dropped from some certain height (and presumably suffer no damage if dropped from below that height). You're then taken to a 100-story building (presumably higher than the certain height) and asked to find the highest floor you can drop a marble from without breaking it as efficiently as possible.

9. Maximising profit - buy sell.

Source: <https://www.geeksforgeeks.org/dsa/best-time-to-buy-and-sell-stock/>

Given an array prices[] of length N, representing the prices of the stocks on different days, the task is to find the maximum profit possible by buying and selling the stocks on different days when at most one transaction is allowed. Here one transaction means 1 buy + 1 Sell.

10. Six balls, two weighing.

Source: [Six Balls, Two Weighings \(cut-the-knot.org\)](http://cut-the-knot.org)

There are six balls of the same shape, but of three colours - two per colour. For every pair of monochromatic balls, one is lighter. All light balls are of the same weight, and so are all the heavy balls. Use 2 weighings on a balance scale to determine which balls are light and which are heavy.

11. Eight balls weight puzzle.

Source: [\(1\) You have eight balls all of the same size. 7 of them weigh the same, and one of them weighs slightly more. How can you find the ball that is heavier by using a balance and only two weighings? - Quora](#)

You have eight balls all the same size. 7 of them weigh the same, and one of them weighs slightly more. How can you find the ball that is heavier by using a balance and only two weighings?

12. 27 coins and two-pan balance.

Source: [Top 10 Puzzles, Riddles, Logical, and Lateral Thinking Questions Asked in Programming Job Interviews \(javarevisited.blogspot.com\)](http://javarevisited.blogspot.com)

There are 27 coins and a two-pan balance. All coins have same weight except for one, which is heavier than all others. All coins look identical. What is the minimum number of weighing required to certainly find out heavier coin?

13. Find the missing number in an array in $O(n)$.

Source: [Find the Missing Number - GeeksforGeeks](#)

Given an array `arr[]` of size **N-1** with integers in the range of **[1, N]**, the task is to find the missing number from the first **N** integers.

Note: There are no duplicates in the list.

Examples: *Input:* `arr[] = {1, 2, 4, 6, 3, 7, 8}`, $N = 7$

Output: 5

Explanation: The missing number between 1 to 8 is 5

14. Equilibrium index of an array in $O(n)$.

Source: [java - Finding an equilibrium index in an int array - Code Review Stack Exchange](#)

A zero-indexed array *A* consisting of *N* integers is given. An equilibrium index of this array is any integer *P* such that $0 \leq P < N$ and the sum of elements of lower indices is equal to the sum of elements of higher indices, i.e.

$$A[0] + A[1] + \dots + A[P-1] = A[P+1] + \dots + A[N-2] + A[N-1].$$

Sum of zero elements is assumed to be equal to 0. This can happen if $P = 0$ or if $P = N-1$. For example, consider the following array *A* consisting of $N = 8$ elements:

$$A[0] = -1 \ A[1] = 3 \ A[2] = -4 \ A[3] = 5 \ A[4] = 1 \ A[5] = -6 \ A[6] = 2 \ A[7] = 1$$

$P = 1$ is an equilibrium index of this array, because:

$$A[0] = -1 = A[2] + A[3] + A[4] + A[5] + A[6] + A[7]$$

$P = 3$ is an equilibrium index of this array, because:

$$A[0] + A[1] + A[2] = -2 = A[4] + A[5] + A[6] + A[7]$$

$P = 7$ is also an equilibrium index, because:

$$A[0] + A[1] + A[2] + A[3] + A[4] + A[5] + A[6] = 0$$

and there are no elements with indices greater than 7.

15. Finding the jar with defective marbles.

Source: [Top 10 Puzzles, Riddles, Logical, and Lateral Thinking Questions Asked in Programming Job Interviews \(javarevisited.blogspot.com\)](#)

You have 10 Jars filled with marbles. Each marble weighs 10 gm, except one Jar which contains defective marbles which weigh 9 gm. Given a scale to weigh, how do you find a Jar with defective marbles in just one measurement? Here the scale is a modern-day electronic scale to measure weight.

16. Measuring nine minutes using sand timer.

Source: [20 Hard Data Science Interview Puzzles which every analyst should know \(analyticsvidhya.com\)](#)

You have two sand timers, which can show 4 minutes and 7 minutes respectively. Use both the sand timers (at a time or one after other or any other combination) and measure a time of 9 minutes.

17. Calculate the number of moves – Hanoi tower.

Source: https://www.transum.org/Maths/Investigation/Tower_Of_Hanoi/

Tower of Hanoi is a mathematical puzzle where we have three rods and n disks. The objective of the puzzle is to move the entire stack to another rod, obeying the following simple rules:

- a. Only one disk can be moved at a time.
- b. Each move consists of taking the upper disk from one of the stacks and placing it on top of another stack i.e. a disk can only be moved if it is the uppermost disk on a stack.
- c. No disk may be placed on top of a smaller disk.

18. The ladder problem – Fibonacci numbers.

Source: [Count ways to reach the n'th stair - GeeksforGeeks](#)

There are n stairs, a person standing at the bottom wants to reach the top. The person can climb either 1 stair or 2 stairs at a time. Count the number of ways, the person can reach the top.

19. Blind game.

Source: [20 Hard Data Science Interview Puzzles which every analyst should know \(analyticsvidhya.com\)](#)

You are in a dark room where a table is kept. There are 50 coins placed on the table, out of which 10 coins are showing tails and 40 coins are showing heads. The task is to divide this set of 50 coins into 2 groups (not necessarily same size) such that both groups have same number of coins showing the tails.

20. Gas stations in a circle.

Source: [Gasoline Stations on a Circular Trek \(cut-the-knot.org\)](#)

Several gas stations on a circular trek have between them just enough gas for one car to make a complete round trip. Prove that if you start at the right station with an empty tank you shall be able to make it all the way around.

21. Detecting a cycle in a singly linked list.

Source: [Loop in a Singly-Linked List » My Tech Interviews](#)

Answer: This is a very typical tech interview question that has a very elegant solution that runs in $O(n)$ time and $O(1)$ space. Also referred to as the “Tortoise and Hare Algorithm”, the solution uses a fast pointer (traversing the list two items at a time) and a slow pointer (traversing the list one item at a time). If there is a loop, then the fast pointer will go around the loop twice as fast as the slow pointer. At one point, these two pointers will point to the same item. Here’s some code to make this clearer.

22. The fruit puzzle.

Source: [Crates of Fruit Puzzle - Solution \(mathsisfun.com\)](#)

You work at a fruit factory.

There are 3 crates in front of you. One crate contains only apples. One crate contains only oranges. The other crate contains both apples and oranges.

And each crate is labelled. One reads "apples", one reads "oranges", and one reads "apples and oranges".

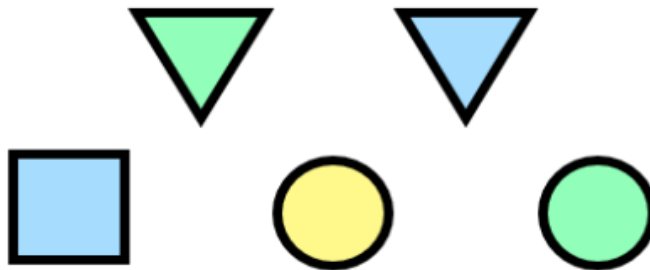
But the labelling machine has gone crazy and is now labelling all boxes incorrectly.

If you can only take out and look at just one of the pieces of fruit from just one of the crates, how can you label ALL of the crates correctly?

23. Oxford University Admissions Question.

Source: <https://mindyourdecisions.com/blog/2021/09/21/a-logic-puzzle-from-an-oxford-university-admissions-question/>

A prize is hidden under one of five objects.



Shilpa is privately told the shape and Colin is privately told the color. Shilpa and Colin are mathematicians who use perfect logical reasoning, and the initial setup is common knowledge to both of them.

The host asks, “Do either of you know where the prize is?” Both of them are silent.

The host then asks, “Do you know now?” Again both of them are silent.

The host asks again, “Do you know now?” At this point both Shilpa and Colin simultaneously say, “Yes!”

Under which object is the prize?